Section 5.7

I. Mean Value Theorem for Derivatives

A. Theorem *Let* f *be continuous on* $[a,b]$ *and differentiable on* (a,b). *Then, there exists a number* $c \in (a,b)$ *such that* $\dfrac{f(b)-f(a)}{b-a} = f'(c)$

II. Mean Value Theorem for Integrals

A. Theorem *Let* f *be continuous on* $[a,b]$. *Then, there exists a number* $c \in (a,b)$ *such that* $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx = f(c)$.

B. Geometric Interpretation

C. Average Value Interpretation